6 Chapter 6: Maxwell's Equations for Time-Varying Fields

6.0 Dynamic Fields

$\vec{E}$ and $\vec{D}$ in the static electric model **are not** related to $\vec{H}$ and $\vec{B}$ in the static magnetic model. In addition, static electric and magnetic fields **do not** give rise to waves that propagate and carry energy and information.

In this chapter, we will see that a **changing** magnetic field induces an electric field, and vice versa; that is, they are coupled, and $\vec{E}$ and $\vec{D}$ will be related to $\vec{H}$ and $\vec{B}$, contrary to their lack of interaction in the static model.

6.1 Faraday's Law

To review, in the static case,

$$\nabla \times \vec{E} = 0.$$

Modifying this equation to cover time-varying fields, in a general sense,

$$\nabla \times \vec{E} = \frac{-\partial \vec{B}}{\partial t}.$$

When $\frac{\partial \vec{B}}{\partial t} = 0$, this relationship simplifies to the static case. Taking the surface integral of both sides of the equation,

$$\int\int_S (\nabla \times \vec{E}) \cdot d\vec{s} = -\int\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}.$$

Using Stoke's theorem to simplify,

$$\oint_C \vec{E} \cdot d\vec{l} = \frac{-d}{dt} \int\int \vec{B} \cdot d\vec{s} = \frac{-d}{dt}\ \Phi.$$

The process described by the above equation is known as **electromagnetic induction**, and the left-hand side can be called the induced voltage, V_{emf}.

Extending to a general case, given N turns in a circuit,

$$V_{\text{emf}} = -N\ \frac{d\Phi}{dt}.$$

In other words, the voltage induced by a changing magnetic field in a closed circuit is proportional to the rate of change in the magnetic flux and the number of turns N in the circuit. Obviously, the induced voltage then induces a corresponding current.

There are different types of induction. 1) stationary circuit/time-varying $\vec{B} \rightarrow$ transformer induction. This is how transformers operate, by using a changing magnetic field to induce a voltage/current in a stationary circuit. 2) Moving circuit/static $\vec{B} \rightarrow$ motional induction. 3) moving circuit/time-changing $\vec{B} \rightarrow$ general case.

Mathetmatically,

$$V_{\text{emf}} = V_{\text{emf}}^{\text{tf}} + V_{\text{emf}}^{\text{m}}.$$

6.2 Stationary Loop in a Time-Varying Field

Reviewing each in more detail, starting with transformer induction, recall that in this type of induction, there will be an induced current generated in a closed loop/circuit due to a changing $\vec{B}$ field. Note that the flux produced by the induced current opposes the change in $\vec{B}$ (Lenz's Law).

Imagine a loop of wire (stationary) through which a changing $\vec{B}$ is passing, and it is growing weaker with time. What direction will the induced current flow? Knowing that the $\vec{B}$ field is decreasing, the induced $\vec{B}$

field will *oppose* this change, and increase. Using the right-hand rule, we can find the direction of the induced current. In the opposite case, where $\vec{B}$ increases, the current will flow in the *opposite* direction.

Example time. A circular loop of N turns of conducting wire lies in the x-y plane with its center at the origin of a magnetic field given by $\vec{B} = \hat{\alpha_z}\ B_0 \cos\left(\frac{\pi r}{2b}\right) \sin(\omega t)$, where b is the radius of the loop and ω is the angular frequency. Given this, find the emf induced in the loop.

First, we can identify the type of induction as transformer induction, since the loop is stationary, and $\vec{B}$ is a function of time t. To find emf,

$$V_{\text{emf}} = -N\frac{d\Phi}{dt}.$$

Therefore, we must first find magnetic flux
Phi as a function of our circuit's structure and the magnetic field $\vec{B}$.

Using the formula for magnetic flux,

$$\Phi = \int\int_S \vec{B} \cdot d\vec{s}.$$

In this case, for a single loop, $d\vec{s} = \hat{\alpha}_z\ r\ dr\ d\phi$.

So,

$$\Phi = \int_0^{2\pi} \int_0^b (\hat{\alpha}_z\ B_0 \cos\left(\frac{\pi r}{2b}\right) \sin(\omega t)) \cdot (\hat{\alpha}_z\ r\ dr\ d\phi).$$

Simplifying using integration by parts,

$$\Phi = 2\pi B_0 \sin(\omega t) \int_0^b \cos\left(\frac{\pi r}{2b}\right) \cdot r \cdot dr = 2\pi B_0 \sin(\omega t)\left(\frac{2b^2}{\pi} - \frac{4b^2}{\pi^2}\right).$$

Taking its time derivative, only the sin function matters, so

$$\frac{d\Phi}{dt} = 2\pi B_0\left(\frac{2b^2}{\pi} - \frac{4b^2}{\pi^2}\right)\omega \cos(\omega t)$$

and so

$$V_{\text{emf}} = -2N\pi B_0\left(\frac{2b^2}{\pi} - \frac{4b^2}{\pi^2}\right)\omega \cos(\omega t)\ \text{V}.$$

Another example problem: an inductor is formed by winding N turns of a thin conducting wire into a circular loop of radius a. The inductor loop is in the x-y plane with its center at the origin, and connected to a resistor R. In the presence of a magnetic field $\vec{B} = B_0(2\hat{\alpha}_y + 3\hat{\alpha}_z)\sin(\omega t)$. Find

a) the magnetic flux linking a single turn of the inductor. In this case, the changing magnetic field has both y and z components, and not all of it will end up as magnetic flux Φ as a result. $d\vec{s}$ is the same as in the prior problem, $d\vec{s} = \hat{\alpha}_z\ (r\ dr\ d\phi)$. Finding magnetic flux using the same integral setup as before,

$$\Phi = \int_0^{2\pi} \int_0^a B_0(2\hat{\alpha}_y + 3\hat{\alpha}_z)\sin(\omega t) \cdot (\hat{\alpha}_z\ (r\ dr\ d\phi)).$$

Simplifying,

$$\Phi = 3B_0\pi a^2 \sin(\omega t).$$